

Y Vectors (P3 & FPI) formulas (dL)

DATE:

Distance between 2 points

x $A\left(\begin{smallmatrix} x \\ y \\ z \end{smallmatrix}\right)$

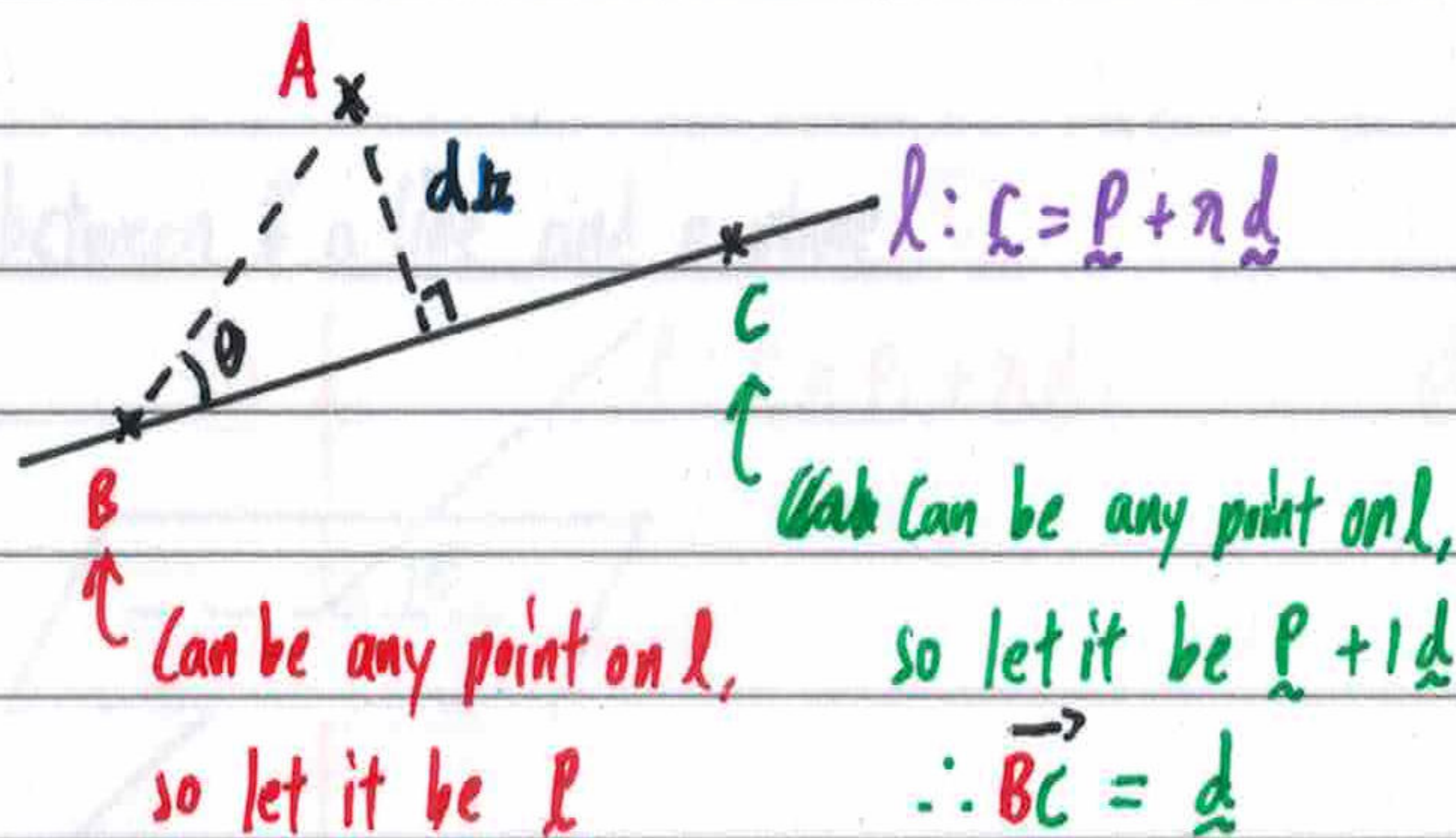
x $B\left(\begin{smallmatrix} x \\ y \\ z \end{smallmatrix}\right)$

$$d = \left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right|$$

$$= \left| \vec{BA} \right|$$

NOTE: dL between 2 parallel lines, 2 parallel planes & a line parallel to a plane, is the same as point-line and point-plane. (Just sub any value to get a point)

Distance between a Point and a line



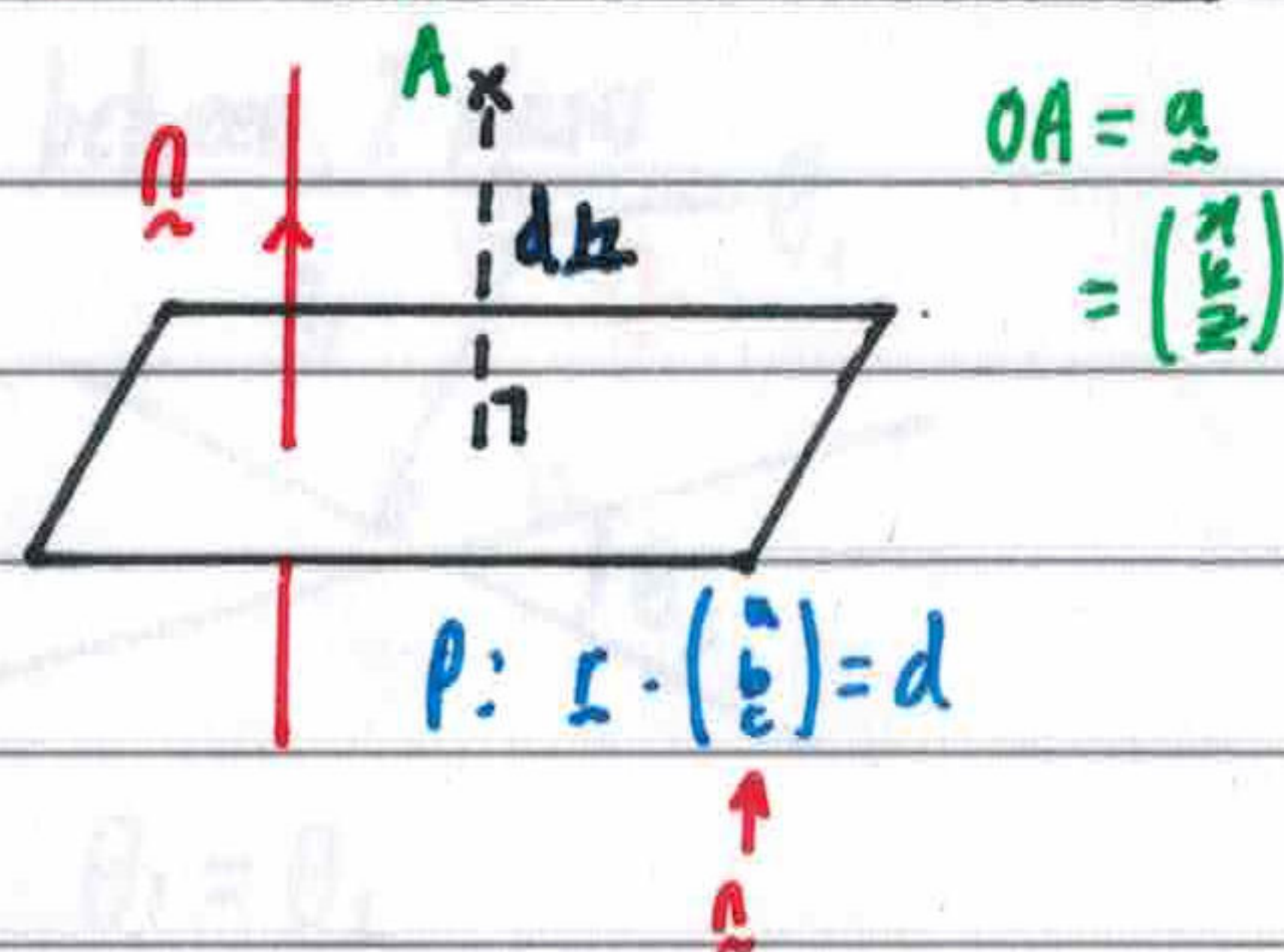
$$dL = \frac{|\vec{BA} \times \vec{BC}|}{|\vec{BC}|}$$

$$= \frac{|\text{hyp} \times \text{adj}|}{|\text{adj}|}$$

$$= \frac{|\text{hyp} \times a|}{|a|}$$

$$= \frac{|\text{Position vector of } l \text{ to point} \times \text{direction vector of } l|}{|\text{direction vector of } l|}$$

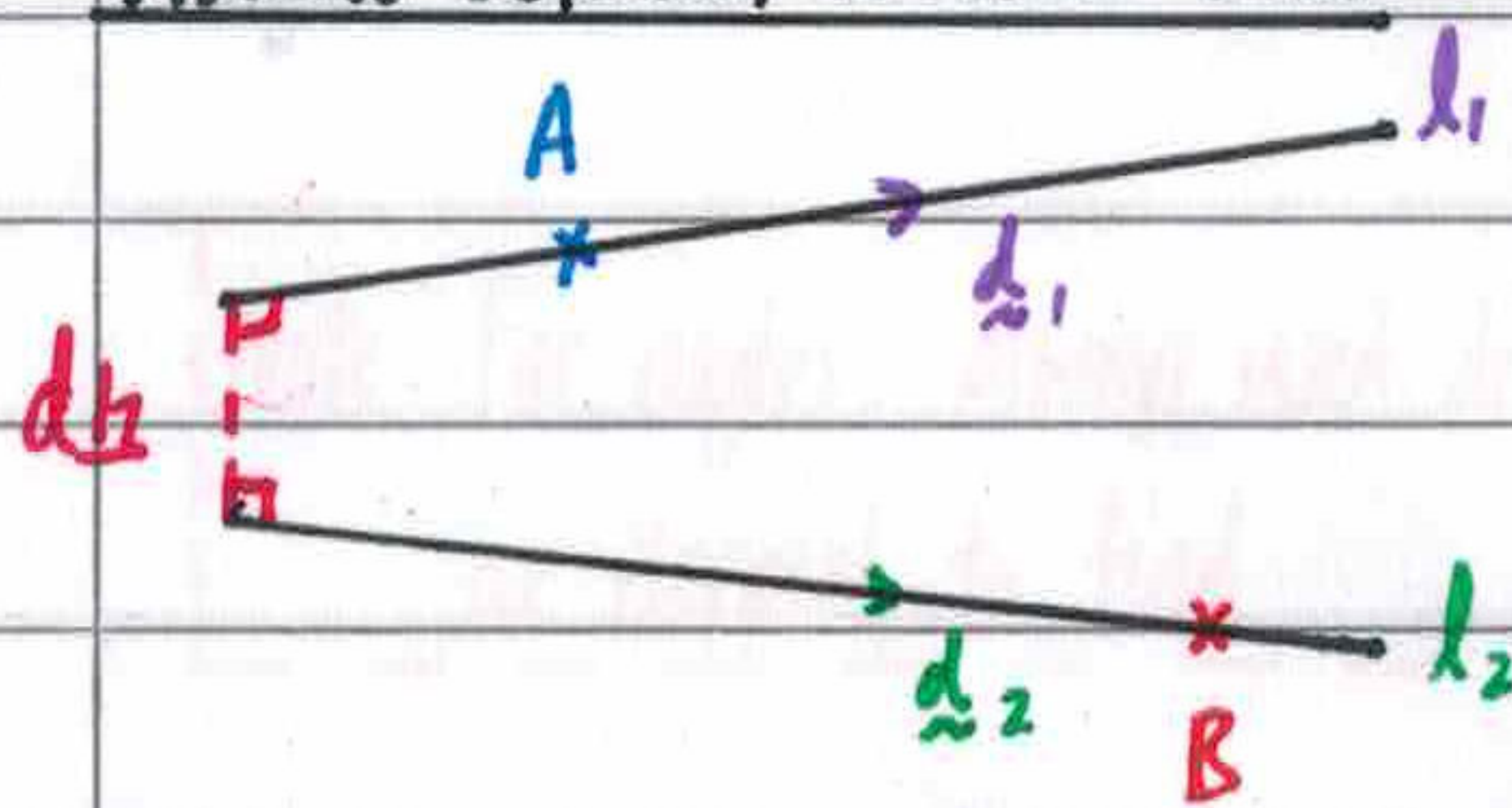
Distance between a Point and a Plane



$$dL = \frac{|ax + by + cz - d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$dL = \frac{|\vec{n} \cdot \vec{a} - d|}{|\vec{n}|}$$

Distance between 2 skew lines



~~AB~~

$$dL = \frac{|\vec{AB} \cdot \vec{d}_1 \times \vec{d}_2|}{|\vec{d}_1 \times \vec{d}_2|}$$

A is any point of l_1

B is any point of l_2

$\vec{d}_1$ is the direction vector of l_1

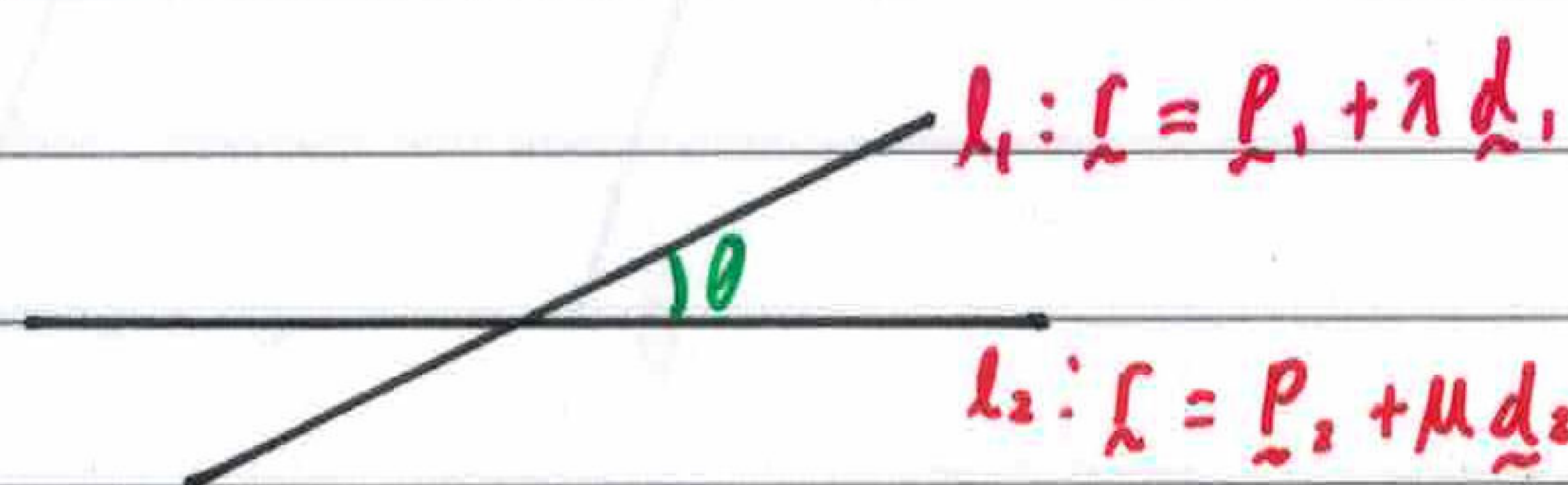
$\vec{d}_2$ is the direction vector of l_2

dL is the perpendicular distance of l_1 & l_2

$$dL = \frac{|\text{vector of any 2 points on both lines} \cdot \text{direction}_1 \times \text{direction}_2|}{|\text{direction}_1 \times \text{direction}_2|}$$

Angle between 2 lines (normally its acute, $180-\theta$ is not)

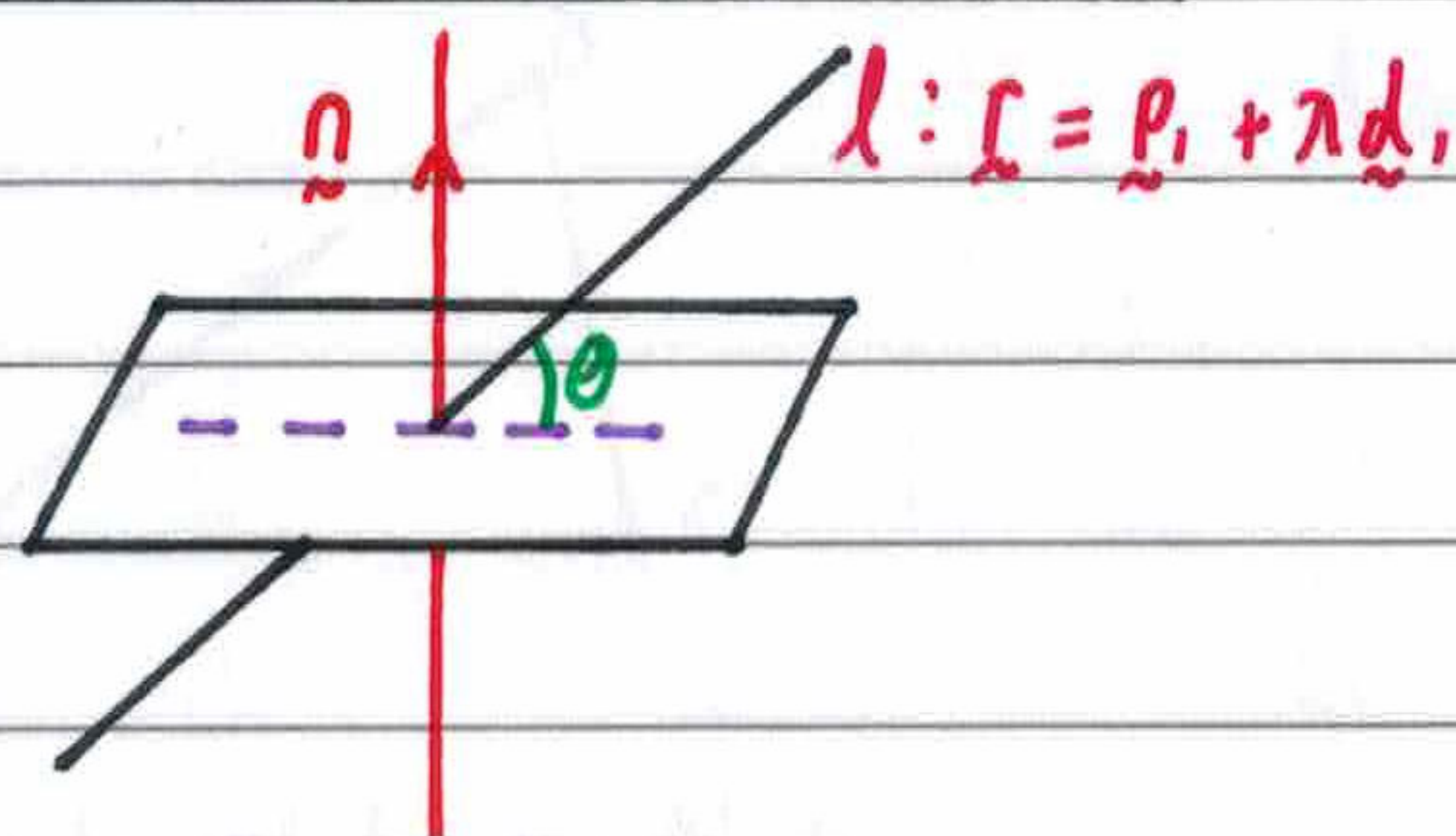
• Lines must intersect, otherwise impossible.



$$\underline{d}_1 \cdot \underline{d}_2 = |\underline{d}_1| |\underline{d}_2| \cos \theta$$

* find the angle between the direction vector.

Angle between a line and a plane



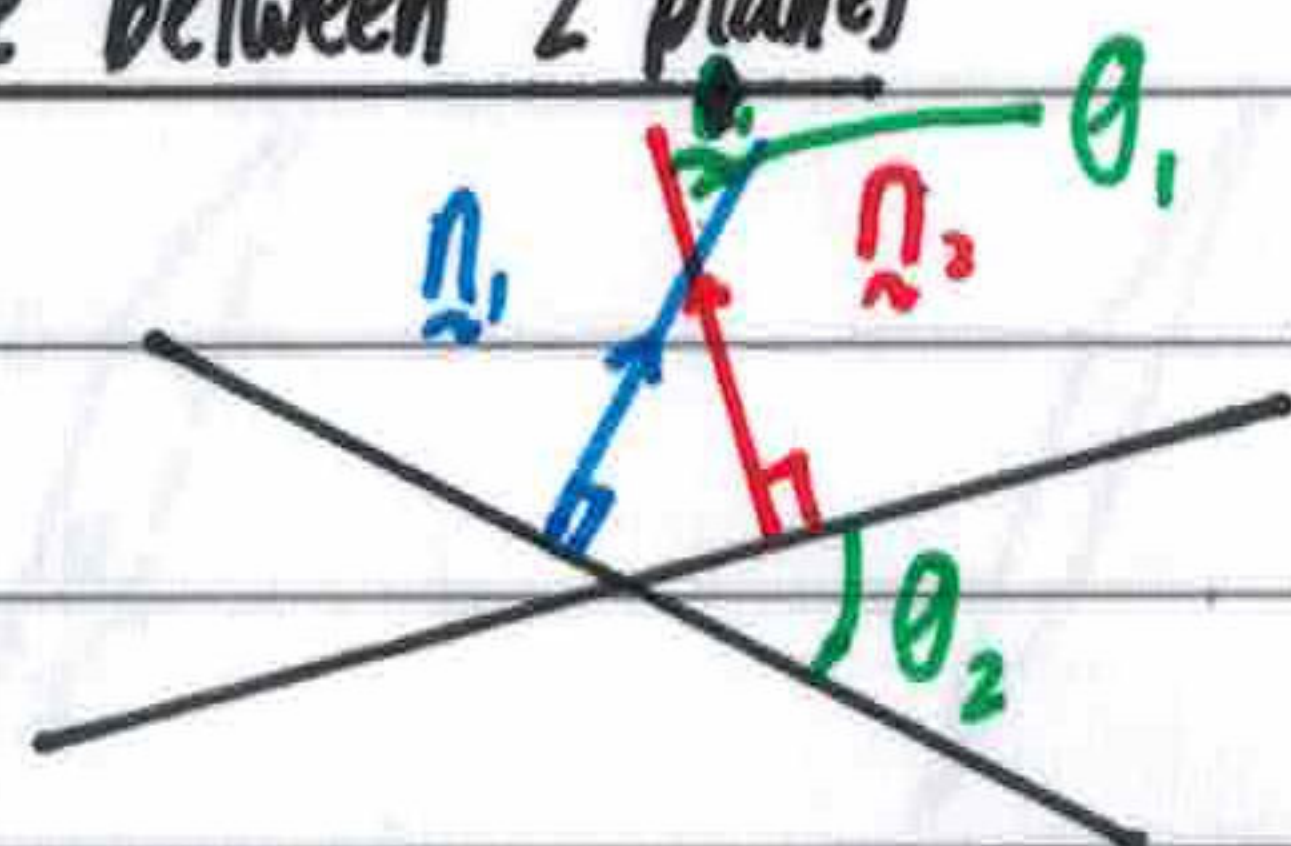
$$\underline{d}_1 \cdot \underline{n} = |\underline{d}_1| |\underline{n}| \cos(90-\theta)$$



$$\underline{d}_1 \cdot \underline{n} = |\underline{d}_1| |\underline{n}| \sin \theta$$

• $\theta = 90 - (\text{angle between line and normal})$

Angle between 2 planes



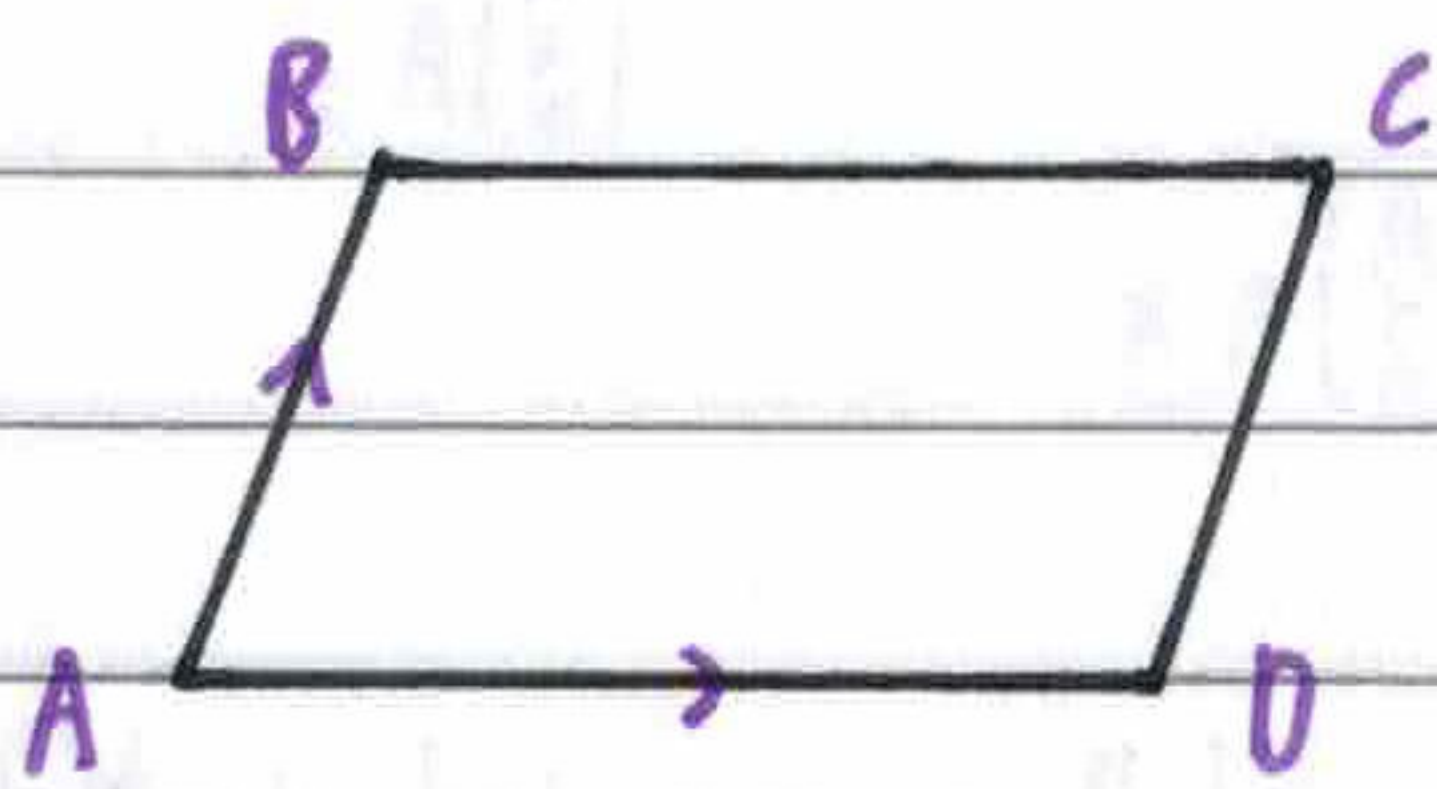
$$\underline{n}_1 \cdot \underline{n}_2 = |\underline{n}_1| |\underline{n}_2| \cos \theta$$

• Angle between normals.

[Note: For angles, always used direction vector or normal to find.]

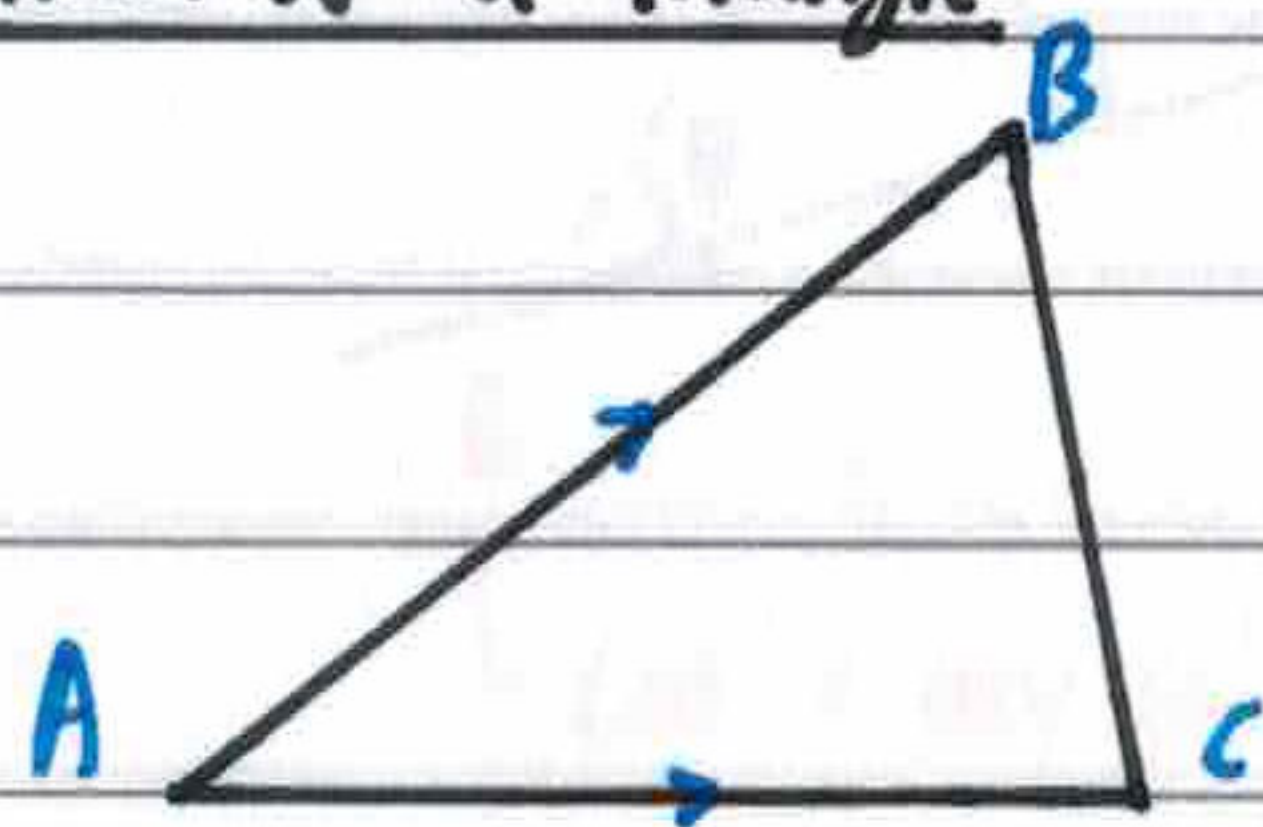
2 Vectors (P3 & FPI) formulas (areas & volumes) DATE:

Area of a Parallelogram



$$\text{Area} = |\vec{AB} \times \vec{AD}|$$

Area of a Triangle

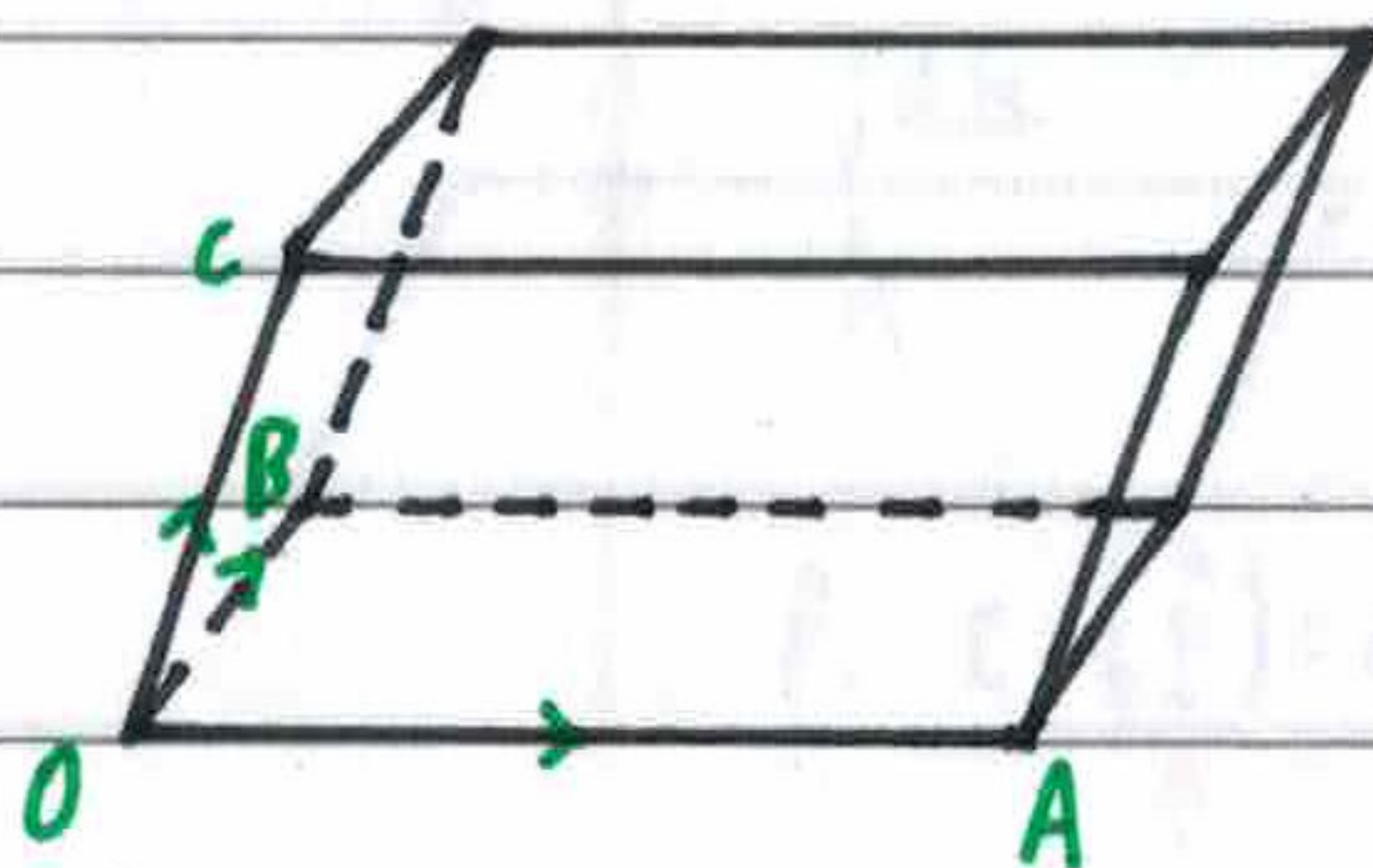


$$\text{Area} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

Note: For areas, it is always |cross product of 2 adjacent vectors|. And only triangle need $\frac{1}{2}$.

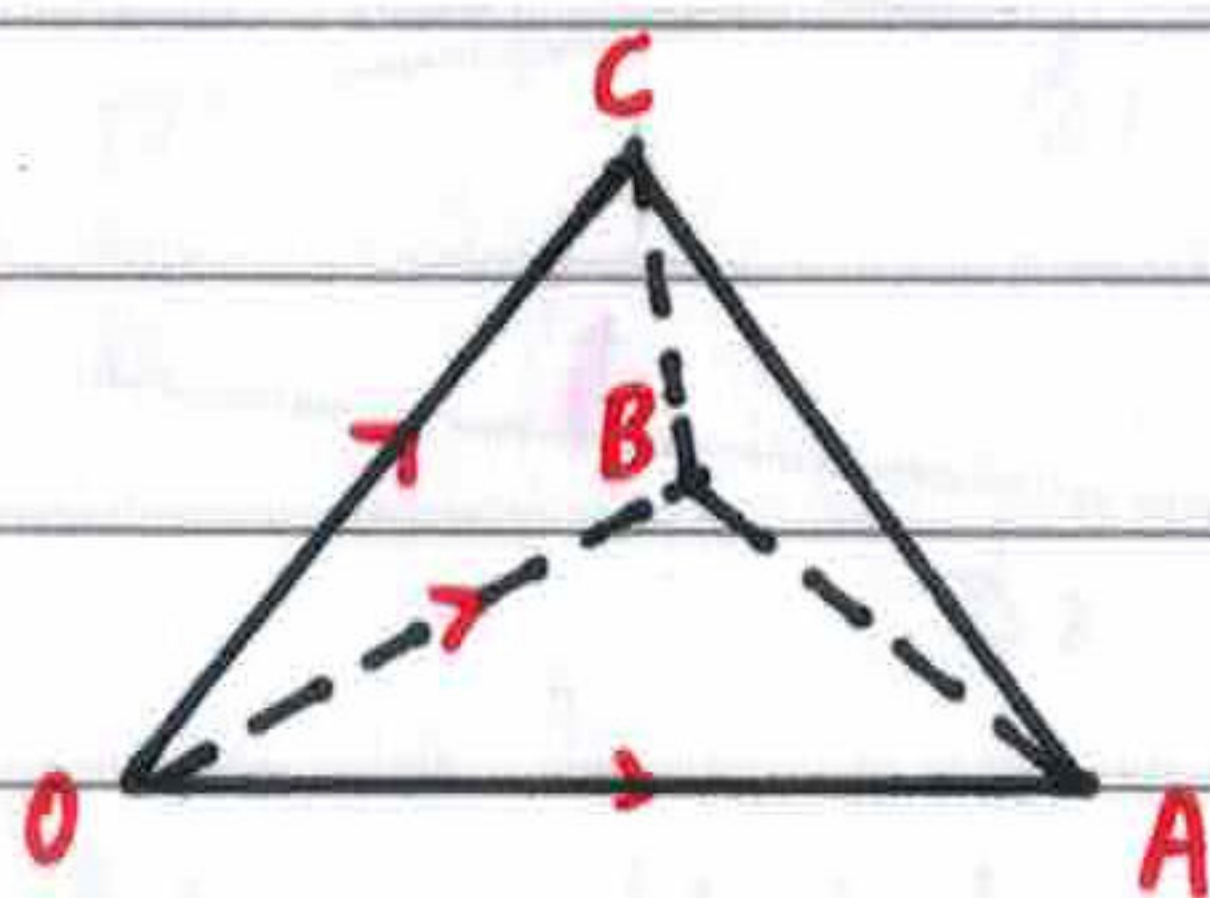
* Can be used for any triangle.

Volume of a Parallelepiped



$$\text{Volume} = |\vec{OA} \times \vec{OB} \cdot \vec{OC}|$$

Volume of a Tetrahedron



$$\text{Volume} = \frac{1}{6} |\vec{OA} \times \vec{OB} \cdot \vec{OC}|$$

Note: For volumes, it is always |cross product of 2 adjacent vectors dot with another vector with direction $\perp$ to the initial|. And only tetrahedron need $\frac{1}{6}$.